

Date _____
Page _____

Experiment :- Study of LAN Topology, network devices and Address configuration.

Objective :-

1. To study the LAN topology used in college campus.
2. To identify networking device & understand their function.
3. To analyze wired & wireless communication method used in the network.
4. To measure speed of network for wired & wireless connections.
5. To obtain Mac, IP Address & Subnet mask.

Problem Statement - 1 :

- Draw & identify LAN Technology of college campus & clearly label all components such as router, switches & etc.

→ Soln :- The college campus network observed this study consist of:

Topology
Address

in collage

understand

communication

or wired

subnet

collage
networks

work

- * Layer-3 core switches in the server room.
- * managed by PoE switches.
- * wireless Access point (APs).
- * Ethernet port in classroom & laboratories.

→ network topology:-

Hybrid Hierarchical Topology which combines:

- 1/ Tree Topology.
- 2/ Extended Star Topology.
- 3/ Ring Topology (server room).

→ The network hierarchical layer:-

- (i) Core layer.
- (ii) Distribution layer.
- (iii) Access layer.

→ Core layer:-

* core layer contains server room

Date _____
Page _____

and server room have. Firewall, Router,
2-L3+ Switches, Patch Rack, Inverter,
etc.

→ Distribution layer :

* Distribution layer contains 4-switches.

* The 4 switches are :-

- ① main Build → R7 (Room no-212, 2nd floor)
- ② main Build → R4 (Room no-301, 3rd floor).
- ③ main Build → R6 (Room no-113, 1st floor).
- ④ main Build → R2 (Room no-201, 2nd floor).

* This 4-switches contain →

- ① 18 Port Gigabit Easy Smart Switch with 16 Port. TL-SG1218MPE
- ② SG3428XMP → 24 Port Gigabit & 4 Port 10 GbE SFP+, L2+ managed switches with 24 Port PoE.

→ Now let's discuss these 2 above components →

① TP-Link

TL-SG1218MP :

- * Likely used for : CCTV camera, wifi access point, etc.
- * 18 Port Gigabit "Easy Smart" Switch
- * 16 Port Support PoE.

② TP-Link Omada SG3428XMP :

- * Higher-end managed layer 2 switch
- * 24 PoE Port, uplinks, 10G SFP.
- * Usually acts as a main distribution for each floor.

→ Access layer :

* Access layer contain 3-switches i.e.

1) main Build R1 - [Room no - 301, 3rd floor]

2) main Build R4 [Room no - 05, Ground floor]

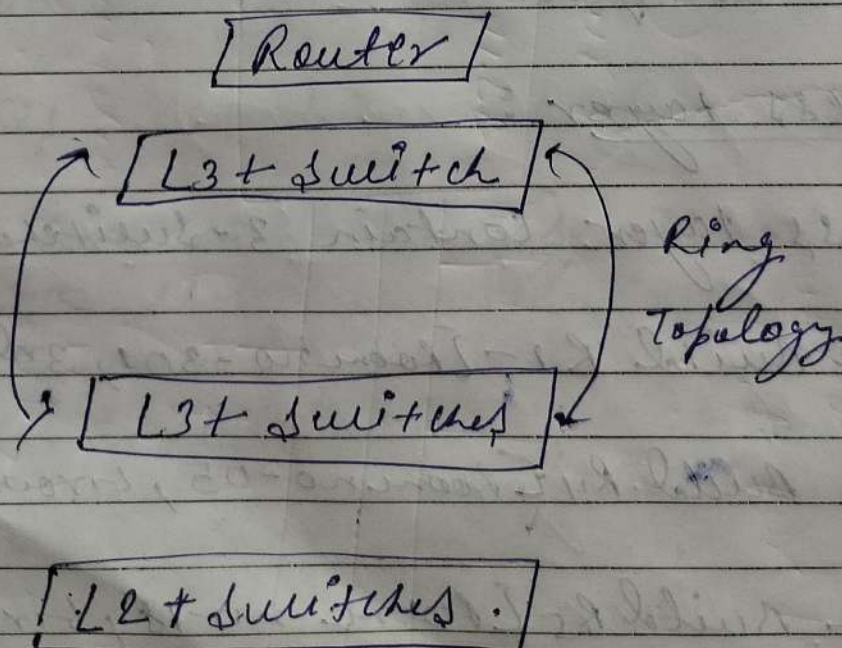
3) main Build R5 [Room no - 18, Ground floor]

* Every switches act as access layer which helps to provide network connectivity to all classroom, lab, faculty room, etc.

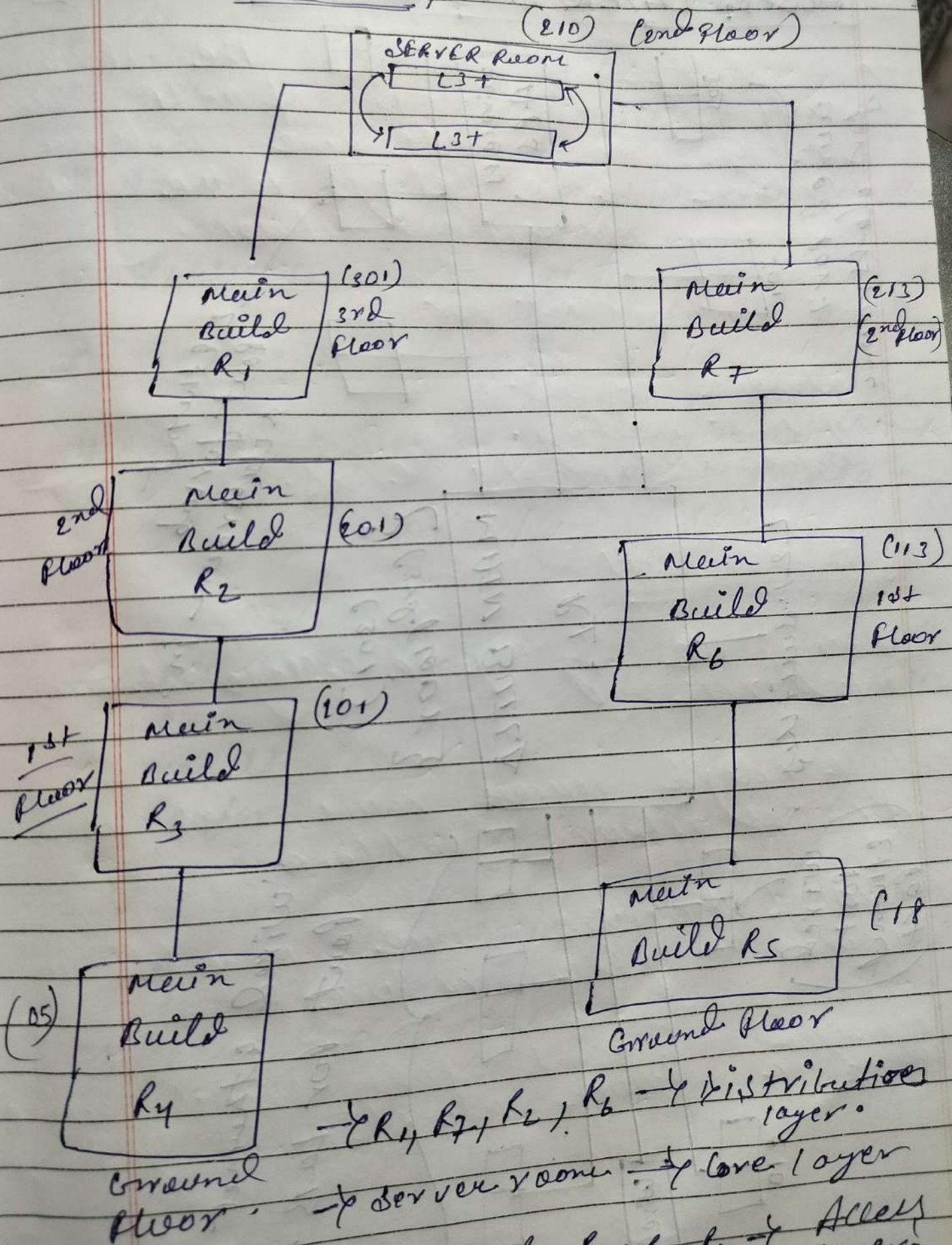
* Access layer directly connect :

- (i) Ethernet port.
- (ii) wireless AP.
- (iii) Lab system.
- (iv) outdoor wireless AP (eu-018).

→ SERVER ROOM Architecture / Topology :-



TOPOLOGY :-

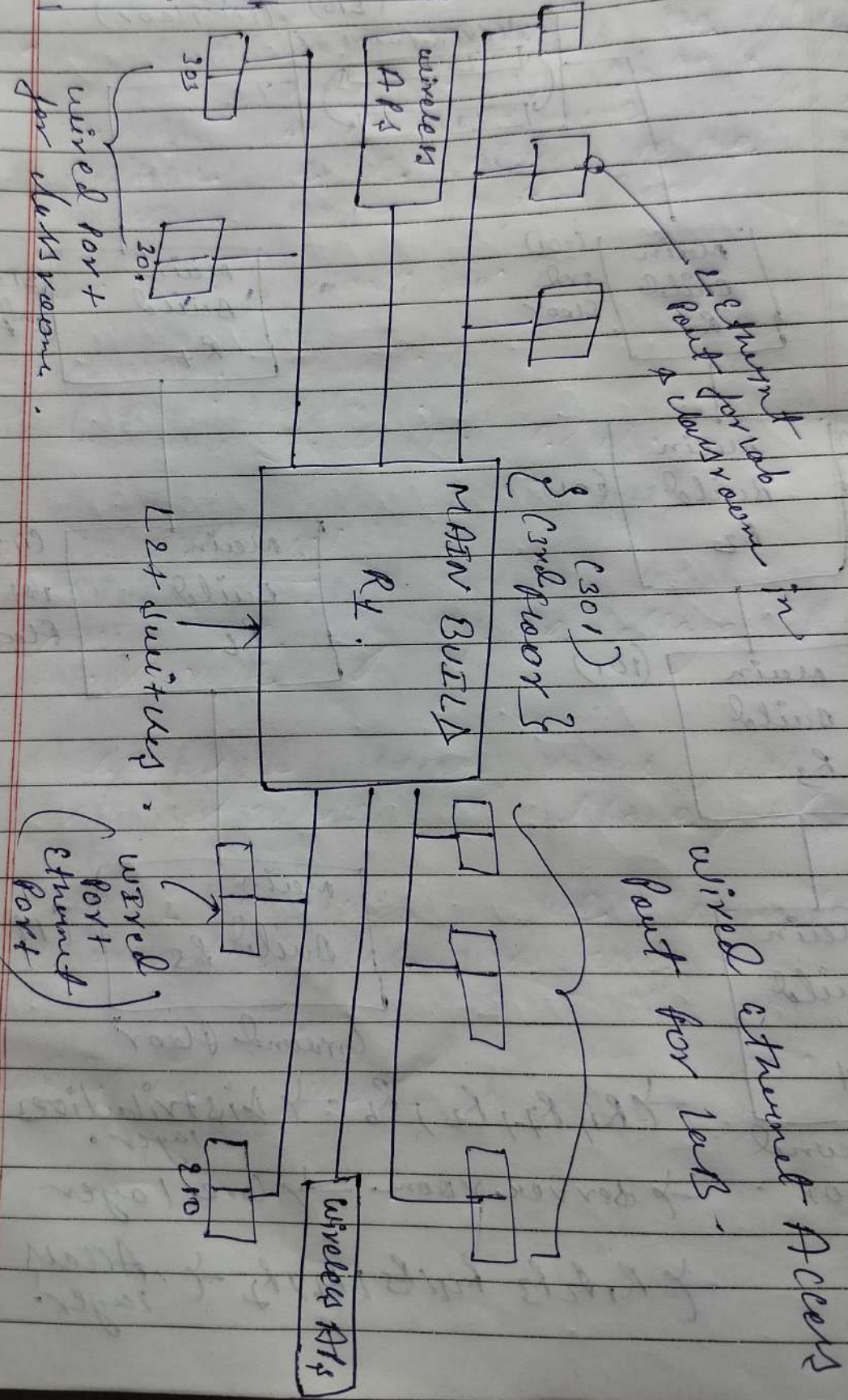


→ R1, R7, R2, R6 → Distribution layer.

→ Server room → Core layer

→ R3, R4, R5, R6, R7 → Access layer.

ACCESS LAYER SWITCH (3rd Floor 301)



→ Difference betⁿ wired & wireless
- speed?

<u>Solⁿ</u>	<u>Wired Network</u>	<u>wireless network</u>
*	Faster Communication	* slower communication.
*	Static connection	* Affected by distance interference.
*	low latency	* higher latency.
*	Dedicated Physical medium.	* shared radio medium.

→ Ethernet provides direct Physical Communication using network cables resulting in higher speed & static performance.

→ wireless connections uses radio waves which are affected by interference, distance, obstacle & multiple connected users.

Q:- Role of Subnet mask in network!

→ A Subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

* It is important for :-

- ① Network Segment.
- ② Efficient routing.
- ③ Communication b/w devices.

→ OBSERVATION :-

During the study of the College network infrastructure :-

- ① A centralized server room was used for as network backbone.
- ② Two L2 switches were connected in redundant cyclic structure.
- ③ Managed TP-Link PoE Switches were used in network cabinets.

* Advantage of Existing Network:

- * Scalable architecture.
- * Centralized management.
- * Support PoE device.
- * Easy expansion of floor & Classroom.
- * Support both wireless & wired.
- * Redundant link for failover.

Conclusion :-

- * The observed Campus LAN follow a hybrid hierarchical topology consist of Tree, extended star Ring. The network uses managed PoE switches wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.

Command Prompt

Microsoft Windows [Version 10.0.26200.8524]
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C:\Users\shiva> ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Local Area Connection* 2:

Media State : Media disconnected
Connection-specific DNS Suffix . :

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
IPv6 Address. : 2409:40e5:11ed:8ae5:dd4a:4d61:1eed:c9fd
Temporary IPv6 Address. : 2409:40e5:11ed:8ae5:fddb:229b:5913:578e
Link-local IPv6 Address : fe80::24b0:cdeb:d354:db73%6
IPv4 Address. : 10.34.33.106
Subnet Mask : 255.255.255.0
Default Gateway : fe80::b844:abff:fe1c:4698%6
10.34.33.111

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :

C:\Users\shiva>

Wi-Fi

throughput

Run new task ...

Intel(R) Wi-Fi 6 AX201 160MHz

500 Kbps

450 Kbps



60 seconds

Send

0 Kbps

Receive

0 Kbps

Adapter name: Wi-Fi
SSID: OnePlus Nord CE 3 Lite 5G
Connection type: 802.11n
IPv4 address: 10.34.33.106
IPv6 address: 2409:40e5:11ed:8ae5:dd4a:4d61:1eed:c9fd
Signal strength:

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Command Prompt

```
Connection-specific DNS Suffix . :  
IPv6 Address. . . . . : 2409:40e5:11ed:8ae5:bd2:5211:4d5b:3d6c  
Temporary IPv6 Address. . . . . : 2409:40e5:11ed:8ae5:f808:5007:a9eb:a052  
Link-local IPv6 Address . . . . . : fe80::d92f:279:e8dd:fd9a%14  
IPv4 Address. . . . . : 10.77.118.154  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::5469:40ff:fe94:9fd7%14  
                             10.77.118.121
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 2:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Wi-Fi:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

C:\Users\shiva>

Ethernet

Throughput

Remote NDIS based Internet Sharing Device

3.6 Mbps

3.6 Mbps

60 seconds

Send

8.0 Kbps

Receive

128 Kbps

Adapter name: Ethernet 2

Connection type: Ethernet

IPv4 address: 10.77.118.154

IPv6 address: 2409:40e5:11ed:8ae5:bd2:5211:4d5b:3d6c



Search



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29-05-2026

 Command Prompt

Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\shiva>gatmac

'gatmac' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\shiva>